No.

Date

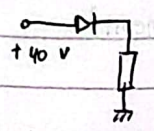
Nama : Devita Syifa Azzahraa

NIM : 2213310009

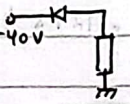
Kelas : 1A TTE

Tugas : Latihan Soal Dioda

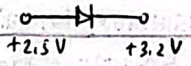
1. Tentukan keadaan masing-masing dioda dengan kondisi bias DC dibawah ini :

a. 

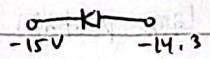
(forward)

b. 

(reverse)

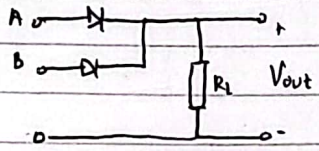
c. 

(reverse)

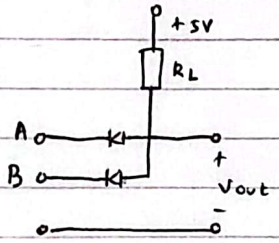
d. 

(forward)

2. Asumsikan dioda ideal. Tentukan logika keluaran tiap rangkaian dengan berbagai kemungkinan tegangan masukan, kemudian tentukan jenis gerbang logikanya :

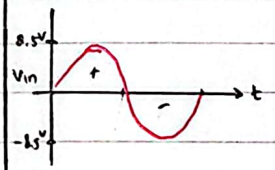
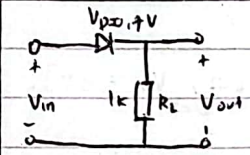
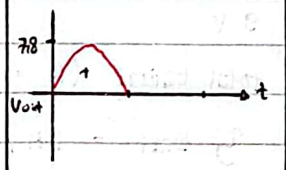
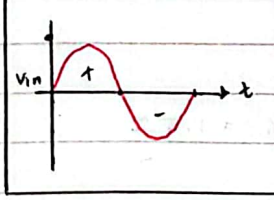
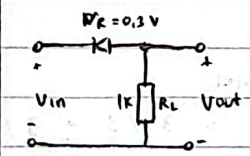
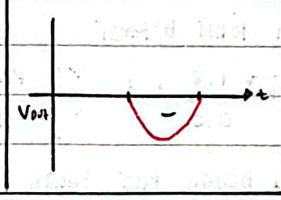
a. 

A(v)	B(v)	Vout(v)	
0	0	0	OR
0	5	5	
5	0	5	
5	5	5	

b. 

A(v)	B(v)	Vout(v)	
0	0	0	AND
0	5	0	
5	0	0	
5	5	5	

3. Gambarkan bentuk gelombang keluaran untuk masing-masing rangkaian dibawah :

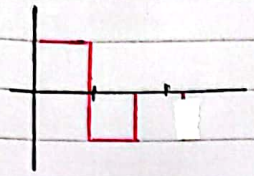
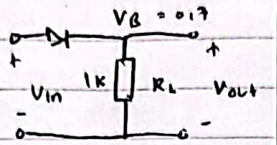
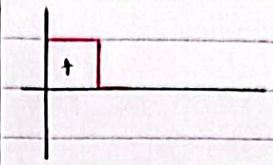
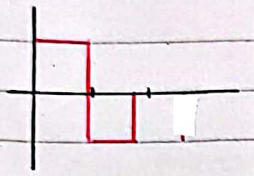
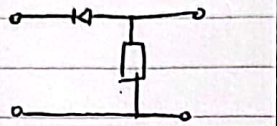
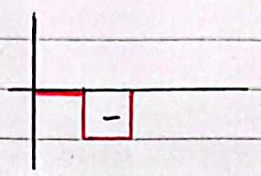
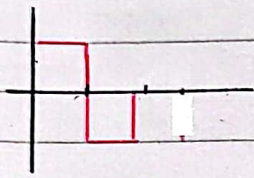
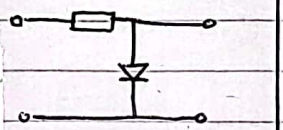
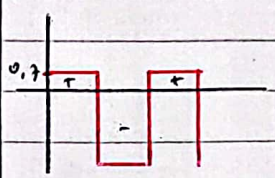
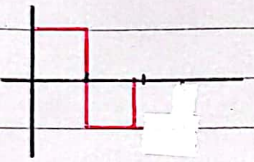
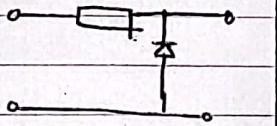
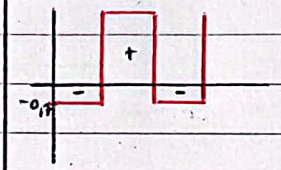
1.	Vinput	Rangkaian Dioda	Voutput
a.			
b.			

Dipindai dengan CamScanner

No.

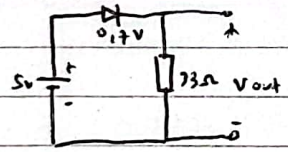
Date . .

4. Asumsikan pendekatan dioda dengan mempunyai tegangan ambang $V_B = 0,7$ V. Gambarkan bentuk gelombang keluaran tiap rangkaian bila gelombang masukannya seperti dibawah :

Vinput	Rangkaian Dioda	Voutput
		
		
		
		

5. Tentukan besarnya IL dan tegangan pada RL bila :

a. Dioda Ideal



$$I_L = \frac{V_S}{33} = \frac{5}{33} = 0,1515 \text{ A}$$

$$V_L = V_S = 5 \text{ V}$$

b. Dik : $V_b = 0,7$ V

$R_L = 33\ \Omega$

$V_{in} = 5$ V

$$I_L = \frac{V_{in} - V_b}{R} = \frac{5 - 0,7}{33} = 0,13 \text{ A}, \quad V_L = V_{in} - V_b = 5 - 0,7 = 4,3 \text{ V}$$

Dipindai dengan CamScanner